RadarMAE: Injecting Physical Inductive Biases into Masked Autoencoders for Advanced Radar Object Detection

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Abstract

Masked Autoencoders (MAE) have emerged as a powerful self-supervised paradigm in vision, yet their direct application to radar perception remains bottlenecked by a fundamental domain gap. Specifically, while conventional MAEs are optimized for RGB images using isotropic grid patching and standard mean squared error (MSE) loss, radar range-azimuth (RA) heatmaps exhibit sinc-oriented anisotropic spatial structures as well as complex physical artifacts, such as heavy-tailed multipath interference and sidelobe spikes. In this paper, we propose **RadarMAE**, the first radar-centric MAE tailored specifically to the physical properties of radar signals. We present an axial tokenization strategy that preserves the resolution disparities and continuous sinc-shaped lobes of RA maps. Furthermore, considering that the range and azimuth dimensions exhibit distinct physical artifacts, we decouple their reconstruction objectives by introducing a multipath distribution loss along the range axis to capture heavy-tailed multipath reflections as well as a sidelobe distribution loss along the azimuth axis to learn deterministic sidelobe spikes. Our extensive experiments demonstrate that injecting these domain-specific physical priors enables robust representation learning for radar object detection. Notably, our method achieves a state-of-the-art 64.17% AP_{BEV} on the K-Radar dataset using only single-frame RA map setting, significantly outperforming the previous baselines.

1 Introduction

Frequency-modulated continuous-wave (FMCW) radar plays a pivotal role in autonomous driving systems, primarily due to its reliable perception capabilities even in adverse weather conditions that severely degrade cameras and LiDAR [1–3]. To extract actionable targets from the radar signals, traditional radar detection algorithms have leveraged signal processing techniques such as the Constant False Alarm Rate (CFAR) detector [4, 5]. However, this approach causes severe information sparsification due to its reliance on a heuristic thresholding pipeline, inevitably discarding weak signal reflections and crucial geometric semantics [6–9]. To overcome this limitation, a paradigm shift toward deep learning-based radar object detection has emerged [1, 2, 6, 10, 11], where supervised learning schemes are generally utilized to capture rich spatial contexts and subtle target signatures directly from raw radar heatmaps. Leveraging these advantages, data-driven radar perception methods can preserve richer signal information and achieve superior detection performance compared to conventional CFAR-based pipelines [2, 6, 12–14].

Despite these advancements, fully unlocking the potential of the radar domain remains bottlenecked by the intrinsic limitations of supervised learning, specifically its strict reliance on massive, high-quality annotations corresponding to input radar heatmaps. Compared to camera or LiDAR data,

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radar heatmaps are inherently non-human-interpretable. This non-interpretability stems from the superposition of target returns with multipath reflections and background clutter, further aggravated by sidelobe artifacts that manifest as ghost replicas of the true response. Such intertwined physical phenomena obscure clear geometric boundaries, thereby making manual annotation exceedingly expensive and error-prone [3, 15]. To address this bottleneck, self-supervised learning (SSL) serves as a key solution [16–19], offering an annotation-free paradigm for extracting robust representations from massive amounts of raw radar data [3, 20–27].

Among the available SSL frameworks [16–19], Masked Autoencoders (MAE) [17] stand out as an exceptionally suitable architecture for object detection in radar heatmaps: its dense patch-level reconstruction forces the model to capture fine-grained local spatial structures, which is crucial for understanding the intricate geometric relationships of object-specific sidelobe and multipath components. Driven by these representational advantages, recent studies have attempted to apply MAE to unlabeled radar signals [20, 21, 26–28]. However, naively adopting vision-centric MAE frameworks yields suboptimal representations. This limitation originates from a fundamental domain gap, which is highly evident when contrasting the camera view with the radar representation, as illustrated in Fig. 1 (a) and (c). While optical images capture clear, surface-level visual boundaries, radar heatmaps render the exact same target as abstract wave propagation properties derived through frequency transformations (*e.g.,* Fourier transform).

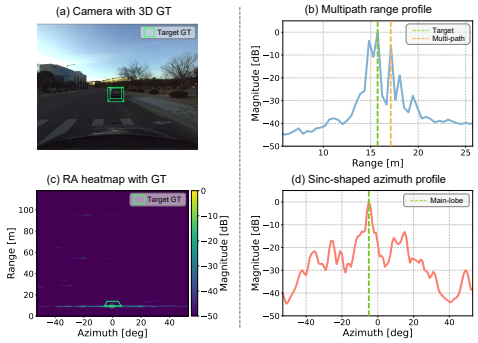


Figure 1: Visual comparison illustrating the vision-radar domain gap. Unlike the clear boundaries in a camera image (a), the corresponding RA heatmap (c) is highly abstract and entangled with inherent physical artifacts. These phenomena are distinctly visible in 1D projections: (b) multipath delays elongating energy along the range profile, and (d) deterministic sinc-shaped spikes along the azimuth profile.

The first mismatch lies in the tokenization strategy. Specifically, the standard MAE utilizes square patches, implicitly assuming an isotropic space where the x (*i.e.,* width) and y (*i.e.,* height) axes possess equivalent spatial semantics. In contrast, radar representation, such as range-azimuth (RA) heatmaps, exhibits intrinsically anisotropic properties. Namely, FMCW radar suffers from severe resolution disparity between the range and angle axes. Moreover, target energy disperses along the entire axes in the form of sinc-shaped lobes in polar coordinates (Fig. 3) [9, 29]. Consequently, the spatial footprint of isotropic square patches in the current MAE scheme is inherently incapable of reflecting this asymmetric nature of radar heatmaps as well as fully capturing the sinc-lobe structures.

The second issue stems from the reconstruction loss in MAE. The standard MAE framework relies on a pixel-wise mean squared error (MSE) loss, which optimizes for estimating the conditional mean and underrepresents extreme statistical deviations [17, 30, 31]. This assumption, however, contradicts the unique physical characteristics of radar heatmaps. The radar signal distribution along the range axis often forms a heavy-tailed distribution due to multipath reflections [32–34], while inherent sidelobes generate continuous chains of sharp peaks along the azimuth axis. Accordingly, when applied to radar data, the MSE objective treats these domain-specific physical artifacts (*i.e.,* multipath tails and sidelobe spikes) as simple noise, aggressively smoothing them out.

To bridge this fundamental domain gap, this paper proposes **RadarMAE**, the first radar-centric MAE framework meticulously designed using physical priors of radar signals in autonomous driving. Our core insight consists of these two novel components: 1) to cover asymmetric and sinc-like target reflections elongated across the entire axes, we propose **axial tokenization**, which simultaneously resolves the severe axis-wise resolution disparities and preserves the spatial continuity of the radar signal; 2) to capture the heavy-tailed multipath components and contiguous sinc-function sidelobes, we propose a **physical distribution-based loss**, which explicitly models the empirical physical properties of each axis and prevents the destruction of critical geometric semantics.

The main contributions of this work are summarized as follows:

1. We propose an axial tokenization strategy that covers the complete information of each axis, which allows the model to explicitly consider the asymmetric characteristics of radar signals and preserve their continuous sinc-function sidelobes.

2. We design a physical distribution-based loss function that improves upon the standard MSE, enabling the model to explicitly preserve the unique physical signatures of radar heatmaps, such as heavy-tailed multipath delays and sinc-shaped sidelobe structures.
3. Using only single-frame RA maps, we establish a new state-of-the-art in radar object detection by achieving a bird's-eye-view (BEV) AP of 64.17% on the K-Radar dataset—a significant 17.85% relative improvement over prior baselines. This result highlights the critical role of physical priors in enabling robust target perception.

2 Related Work

Radar Perception with SSL. Unlike camera or LiDAR data, radar signals inherently lack human interpretability, making it exceedingly difficult to construct high-quality, large-scale labeled datasets [3, 15]. To circumvent this critical limitation, recent research has increasingly gravitated toward SSL to exploit vast amounts of unannotated raw radar data [3, 20–24, 26, 27]. Early SSL frameworks in this domain primarily adapted contrastive learning (CL) paradigms, such as SimCLR [16] and MoCo [18], to radar heatmaps [3, 24, 25, 35]. However, CL is fundamentally designed to extract global, instance-level representations; while advantageous for classification, this high-level abstraction obscures fine-grained spatial details and yields suboptimal representations for dense prediction tasks such as object detection, where precise localization is critical [36, 37]. In contrast, MAE [17] formulates dense patch-level reconstruction as a pretext task, compelling the network to recover fine-grained local structures rather than instance-level abstractions [38, 39]. This property aligns naturally with radar heatmap-based detection, where target-specific geometric cues—such as sidelobes and multipath interference—are encoded in delicate spatial patterns that global instance-level contrastive objectives may fail to explicitly preserve [29, 32, 36, 37].

Motivated by the representational advantages of masked modeling, recent work has actively extended MAE to various radar applications. In autonomous driving, frameworks such as Robu-MARC [26] have utilized masked autoencoding to enhance radar-camera fusion and robust 3D perception, while BEV-MAE [27] adopts BEV-guided masking for LiDAR point cloud pre-training and 3D object detection. In the remote sensing domain, models like CROMA [20] and SARMAE [28] have adopted patch-based masking to extract robust representations from multi-modal and SAR imagery. While these pioneering efforts establish the potential of MAE in the radar domain, their primary focus remains confined to application-level usage rather than fundamentally redesigning the MAE framework itself. Namely, existing radar MAE approaches directly adopt the core mechanics of vision-centric MAE (*i.e.,* tokenization scheme and pixel-wise losses) without radar-specific adaptations. Consequently, the learned representations cannot reflect the anisotropic geometry and dispersion of sinc-shaped sidelobes that fundamentally characterize radar data, leaving substantial room for radar-specific designs.

Injecting Domain-Specific Physical Priors into SSL. Incorporating domain-specific physical priors has become a prevailing strategy for representation learning on signals that inherently deviate from the standard image modality. For instance, in audio and spectrogram analysis, frameworks such as AudioMAE [40] and AST [41] improve representation learning by introducing asymmetric patches tailored to the distinct physical semantics of the time and frequency axes. Similarly, adaptations such as VideoMAE [42] for video and SatMAE [43] for remote sensing incorporate structural priors specific to their respective domains. These successes across diverse modalities collectively demonstrate that injecting the inherent heterogeneity of signal dimensions is crucial for capturing robust features. Despite this prevailing trend across adjacent modalities, incorporating analogous radar-specific inductive biases remains largely unexplored—a gap that our work directly targets.

3 Preliminary

3.1 Range-Azimuth Representation in FMCW Radar

Physical Basis of the Radar Domain Gap. While camera images capture dense and isotropic spatial reflections in a single snapshot, FMCW radar RA heatmaps are constructed through two orthogonal and physically distinct signal processing pipelines [44, 45]. The range axis is the mapped result of the round-trip time delay of chirp signals reflected from objects. Consequently, the spatial resolution along this axis is fundamentally determined by the sweep bandwidth of the transmitted

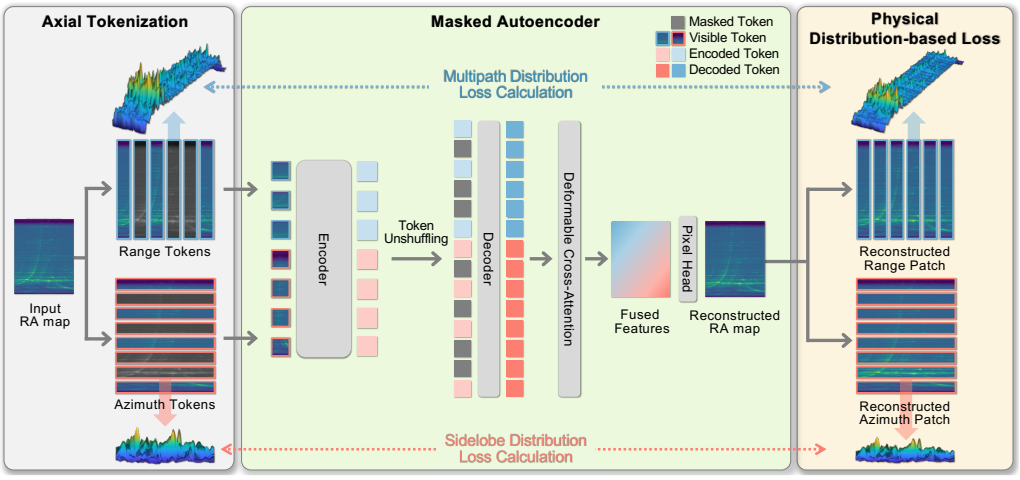


Figure 2: Overall pre-training pipeline of RadarMAE. The input RA map undergoes axial tokenization to preserve the contiguous geometry of sinc lobes. Following intensity-weighted masking, visible tokens are encoded and decoded using deformable cross-attention for feature fusion. Finally, the reconstructed patches are optimized via a physical distribution-based loss to explicitly capture multipath delays and sidelobe spikes.

signal. Conversely, the azimuth axis is derived from the spatial phase differences of the reflected signals received across multiple antenna elements. Accordingly, the angular resolution along this axis is fundamentally determined by the size of the radar's physical or virtual antenna aperture. Originating from fundamentally different physical mechanisms (bandwidth vs. spatial aperture), these two dimensions intrinsically embed highly distinct, dimension-specific physical artifacts into the resulting heatmap, which we explicitly formulate in the following subsection.

3.2 Physical Artifacts in RA Heatmap Distributions

Resolution Disparity. Driven by the aforementioned hardware limitations, the fundamentally different physical mechanisms defining range and azimuth yield severe resolution asymmetry. This structural disparity results in highly anisotropic target profiles across the two dimensions, as can also be observed in Fig. 3.

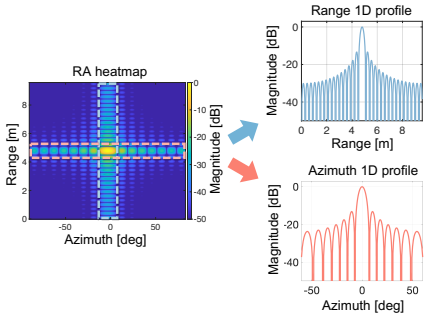


Figure 3: Ideal sinc-shaped point spread sidelobes of the FMCW radar RA heatmap.

Multipath Delay. The range dimension exhibits its own distinct artifact: multipath delay, which significantly distorts the range distribution (Fig. 1b) [32–34]. This phenomenon is primarily driven by multiple reflections between the vehicle's undercarriage and the ground surface. Unlike stationary clutter (*e.g.,* roadside infrastructure), which typically returns simple direct reflections, the geometry beneath a vehicle induces complex, delayed indirect paths. Because these ground-bounced paths cover longer physical distances, the signal energy is systematically mapped to farther range bins, forming an asymmetric, heavy-tailed profile [32–34]. Crucially, since these multipath artifacts form ghost reflections of targets, understanding multipath is crucial for robust radar embedding. Furthermore, this heavy tail acts as a discriminative physical signature that inherently separates vehicles from stationary clutter, providing vital contextual hints for object detection.

Sinc-Shaped Sidelobes. In radar signal processing, applying the fast Fourier transform (FFT) over finite boundaries—restricted by limited bandwidth in range and finite antenna arrays in azimuth—mathematically acts as a rectangular window. This inevitably causes spectral leakage in both dimensions, dispersing a target's energy and forming sidelobes governed by $\mathrm{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$ [29, 45, 46]. Similar to multipath effects, these sinc-shaped artifacts form multiple ghost reflections around the target (Fig. 3). Nevertheless, because these sidelobes manifest as deterministic peaks

rather than mere random noise, they can aid in understanding the target's underlying properties. Therefore, these inherent sinc characteristics must be explicitly considered when forming effective radar representations.

4 Proposed Methods

To address the inherent limitations of applying standard MAE directly to the radar domain, RadarMAE introduces a fundamentally improved architecture designed to explicitly learn the physical semantics of radar signals from RA heatmaps. As depicted in Fig. 2, the model is first pre-trained to reconstruct masked signals, thereby learning the rich physical priors of the radar domain. After pre-training, the decoder is discarded, and a detection head is appended to the pre-trained encoder to fine-tune the network for the downstream task of object detection. The details of each component are as follows.

4.1 Axial Tokenization

Axial Tokenization. Standard MAE architectures adopt isotropic square patches, which intrinsically fail not only to accommodate the severe resolution disparities between the range and azimuth axes, but also to cover the sinc-shaped reflections spanning across the entire axes. To resolve this geometric mismatch, we introduce a simple but powerful axial tokenization strategy, which disentangles the target signal into *orthogonal axial stripes*. As illustrated in Fig. 2, by spanning the entire length of each respective axis, these stripes inherently adapt to the extreme resolution disparities while simultaneously preserving the complete spatial continuity of extended radar signatures without fragmentation. Prior to linear projection, we apply a lightweight per-stripe early convolution to stabilize local feature extraction while strictly preventing inter-stripe information leakage. The resulting feature maps ($\tilde{\mathbf{x}}_i^r$ and $\tilde{\mathbf{x}}_j^a$) are then independently projected into range-wise and azimuth-wise tokens:

$$\mathbf{r}_i = W_r \cdot \mathrm{vec}(\tilde{\mathbf{x}}_i^r) + \mathbf{b}_r \in \mathbb{R}^d, \tag{1}$$

$$\mathbf{a}_j = W_a \cdot \mathrm{vec}(\tilde{\mathbf{x}}_j^a) + \mathbf{b}_a \in \mathbb{R}^d. \tag{2}$$

Here, $\mathrm{vec}(\cdot)$ denotes the vectorization operation that flattens the input stripe, d is the hidden embedding dimension, and W and $\mathbf{b}$ represent the learnable weight matrices and bias vectors. By decoupling the axes, this formulation reduces the token sequence length from the standard $\mathcal{O}(\frac{H \cdot W}{P_r \cdot P_a})$ to $\mathcal{O}(\frac{H}{P_r} + \frac{W}{P_a})$, achieving highly efficient computation. To explicitly inject radar geometry, we replace conventional index-based embeddings with a physical positional encoding (PPE) derived from absolute sensor coordinates. After adding the PPE and learnable axis-type embeddings, the orthogonal sequences are concatenated into a unified input $\mathbf{Z}_0 = [\mathbf{r}; \mathbf{a}]$.

Masking Strategies. While axial tokenization is highly effective for capturing extended sinc signatures, its orthogonal intersecting structure introduces a critical issue: the potential for local information leakage where the range and azimuth stripes overlap each other. To resolve this and prevent the model from relying on these shortcuts, we explicitly break this spatial continuity by occluding contiguous segments rather than isolated patches. Building upon this, we further introduce an intensity-weighted block masking strategy specifically designed for sparse radar RA heatmaps. While standard random masking is highly effective for densely distributed camera images, applying it to radar RA heatmaps risks entirely erasing rare target signals, causing the network to collapse into trivial background interpolation. Based on the domain prior that high-intensity reflections contain informative target signatures, we prioritize masking these contiguous high-intensity regions to force the network to reconstruct meaningful radar responses rather than background patterns. Detailed visualizations and ablation analyses of this masking strategy are provided in Appendix F. Finally, to explicitly preserve their distinct physical properties throughout the pipeline, the masked positions are replaced with dedicated learnable mask tokens ($\mathbf{m}_R$, $\mathbf{m}_A$) for the range and azimuth axes, respectively.

4.2 RadarMAE Architecture

Encoder and Decoder. Following the asymmetric MAE paradigm [17], our encoder processes only the visible subset of the unified token sequence $\mathbf{Z}_0$, while a lightweight 8-block decoder reconstructs the signal. Uniquely, processing these concatenated orthogonal axial stripes through a

shared Transformer encoder allows distinct axial inputs to deeply interact via self-attention. This mechanism learns critical cross-axis coupling and geometric dependencies from highly sparse data. To initiate decoding, we pad the sequence with dedicated learnable mask tokens ($\mathbf{m}_R$ and $\mathbf{m}_A$) tailored to the range and azimuth axes to preserve their distinct physical properties. This overall asymmetric design compels the model to extract robust semantic representations of the radar space rather than merely memorizing high-fidelity continuous signals.

Deformable Cross-Attention-Based Fusion. To reconstruct the decoupled axial tokens back into a dense 2D patch grid, a robust geometric mapping is required. We introduce an axial deformable cross-attention mechanism [47]. Let $\mathbf{Q} \in \mathbb{R}^{H_g \times W_g \times d}$ denote a set of learnable 2D spatial queries assigned to a uniform reference grid, where H_g and W_g represent the spatial dimensions. For a query $\mathbf{q}_{ij}$ at reference point p_{ij}, the network predicts continuous 2D sampling offsets Δp_{ijk} and attention weights A_{ijk} for K sampling points. The projected sampling location $p_{ij} + \Delta p_{ijk}$ is subsequently decomposed into orthogonal coordinates x and y. Using these coordinates, the model independently extracts continuous features from the range token bank $\mathbf{V}_r$ and azimuth token bank $\mathbf{V}_a$, which are derived from the outputs of the preceding decoder blocks, via bilinear grid sampling. These sampled features are weighted by A_{ijk}, aggregated along their respective axes, and finally fused via element-wise averaging to yield the 2D feature map. This strategy enables the 2D queries to adaptively integrate long-range geometric cues from orthogonal axial tokens.

4.3 Physical Distribution-Based Loss

Standard pixel-wise reconstruction objectives, such as the MSE used in conventional MAEs, fundamentally optimize for conditional pixel averages [17, 30, 31]. Consequently, they tend to over-smooth the unique physical distributions inherent in radar data, treating target deviations as uniform noise and failing to capture highly heterogeneous, axis-specific artifacts. To overcome this limitation, rather than merely minimizing pixel-wise distances, we introduce physical distribution-based losses that explicitly enforce the preservation of these intra-stripe distributional properties. Accordingly, the overall pre-training objective of RadarMAE is formulated by augmenting the base stripe reconstruction loss ($\mathcal{L}_{rec}$) with our novel physical distribution-based losses tailored for each axis:

$$\mathcal{L}_{total} = \mathcal{L}_{rec} + \lambda_A \mathcal{L}_A + \lambda_R \mathcal{L}_R. \tag{3}$$

Here, $\mathcal{L}_{rec}$ denotes the base intensity-weighted mean absolute error (L1) loss for stripe reconstruction. By defining high-intensity regions within a stripe as critical target reflections in the driving environment, this weighting strategy explicitly penalizes errors in these areas, forcing the network to focus on reconstructing essential target responses rather than background noise. Meanwhile, $\mathcal{L}_A$ and $\mathcal{L}_R$ are the explicitly formulated distribution matching constraints for the azimuth and range axes, respectively. The hyperparameters λ_A and λ_R balance the contribution of each physical prior. The detailed formulations of these dimension-specific physical losses are described as follows.

Sidelobe Distribution Loss. As depicted in Fig. 3, the finite aperture of the antenna array inevitably induces deterministic, sinc-shaped sidelobes along the azimuth axis [45, 48]. Rather than smooth gaussian noise, these artifacts manifest as periodic, sharp spikes with complex spatial extents. Statistically, capturing this precise geometric signature requires aligning multiple statistical moments. While the variance (σ^2) governs the overall spatial dispersion and beamwidth of the lobe energy, the skewness (g_1) accounts for structural asymmetries often caused by overlapping target reflections or off-center scattering. Finally, the highly leptokurtic nature of the sharp peaks is explicitly characterized by an exceptionally large kurtosis (g_2). To ensure the network faithfully reconstructs these high-frequency geometric signatures without blurring them into the background, we propose a higher-order moment matching loss for the masked azimuth tokens $\mathbf{m}_A$. Specifically, we explicitly align the variance, skewness, and kurtosis of the predicted distribution $\hat{y}$ with the ground-truth y:

$$\mathcal{L}_A = |\sigma^2(y) - \sigma^2(\hat{y})| + |g_1(y) - g_1(\hat{y})| + \lambda_k |g_2(y) - g_2(\hat{y})|. \tag{4}$$

This calibrated moment matching empowers the model to accurately capture the exact morphology of the sinc-shaped sidelobes, providing robust spatial features for object orientation and size estimation.

Multipath Distribution Loss. As formulated in Sec. 3.2, radar waves frequently undergo indirect reflections, such as those from vehicle undercarriages or surrounding structures [32–34]. These delayed reflections generate a heavy-tailed intensity distribution along the range axis [32–34], which implicitly encodes useful geometric and material cues [49, 50]. Suppressing these physical heavy tails can therefore remove contextual information required for accurate localization. To explicitly preserve

Table 1: Tokenization ablation. For all configurations, MSE loss, random masking, and 500 epochs are applied identically. GFLOPs are measured during pre-training. Best results are shown in bold.

Patch size	# Tokens	$AP_{\mathrm{BEV}}(\%)\uparrow$	Encoder GFLOPs↓	Total GFLOPs↓
16×16	112	49.46	**17.67**	**23.97**
8×8	448	54.91	48.51	75.47
4×16	448	55.23	50.04	77.93
16×4	448	54.59	47.65	74.15
Axial	142	**57.46**	22.22	30.14

this domain-specific distribution, we introduce a 1D Wasserstein-based distribution loss tailored to the range axis [51]. We adopt the 1D Wasserstein distance instead of traditional divergence-based metrics (*e.g.,* KL or JS divergence) due to its favorable optimization behavior. When the predicted and target distributions have little or no support overlap, KL divergence may become undefined or unbounded, while JS divergence can saturate and provide weak gradients for distribution alignment. In contrast, the Wasserstein distance provides a meaningful geometric discrepancy and a smooth optimization signal even for disjoint supports. Thus, the exact 1D Wasserstein distance can be efficiently computed as the mean absolute difference between the sorted samples:

$$\mathcal{L}_R = \frac{1}{|\mathbf{m}_R|} \sum_{i=1}^{|\mathbf{m}_R|} \left| \mathrm{sort}(y)_i - \mathrm{sort}(\hat{y})_i \right|. \tag{5}$$

Here, $\mathbf{m}_R$ represents the set of masked range tokens, while y and $\hat{y}$ denote the ground-truth and predicted scalar intensities, respectively. The $\mathrm{sort}(\cdot)$ operator is differentiable almost everywhere and has a computational complexity of $\mathcal{O}(N \log N)$, introducing negligible overhead to the training process.

4.4 Fine-Tuning for Object Detection

After pre-training, we discard the decoder and fine-tune the encoder with an anchor-based detection head identical to RTNH [2]. Axial tokens are projected to a BEV feature map through a lightweight stripe-to-BEV mapping, followed by ASPP-Lite [52] for multi-scale context.

5 Experiments

5.1 Datasets and Experimental Settings

K-Radar Dataset. The K-Radar dataset is a large-scale, widely used 4D radar dataset for 3D object detection in autonomous driving, providing 4D radar tensors (4DRT) that contain power measurements along the Doppler, range, azimuth, and elevation dimensions [2]. In this work, we generate RA maps from the provided 3D range-azimuth-elevation (RAE) maps by averaging over the elevation dimension, and use the resulting RA maps as the sole input modality for both RadarMAE pre-training and downstream object detection. We focus on the sedan class within a longitudinal range of $[0, 72]\,\mathrm{m}$ and a lateral range of $[-6.4, 6.4]\,\mathrm{m}$.

Evaluation Metrics. We evaluate detection performance using the intersection-over-union (IoU)-based average precision (AP) under the standard KITTI protocol [53] with 40 recall positions. We report the AP for both BEV predictions (AP_{BEV}) and 3D bounding box predictions (AP_{3D}) on the sedan class, at IoU thresholds of 0.3 and 0.5 to assess the spatial localization accuracy of the model.

5.2 Effect of Axial Tokenization in MAE

To analyze the effect of the tokenization scheme in MAE, we first conducted a series of experiments with varying patch sizes, as shown in Table 1. Decreasing the isotropic square patch size from 16×16 to 8×8 yields a substantial improvement in AP_{BEV}. This indicates that finer resolution can better capture the physical signatures of sparse radar signals, albeit at the cost of an increased computational burden. To accommodate the distinct resolution disparities across the RA map axes and to encompass wider sidelobe regions within a single token, we applied asymmetric 4×16

Table 2: Comparison on K-Radar-20. BEV and 3D AP are reported at IoU 0.3/0.5. Methods marked with * use adapted backbones for the corresponding radar input. Best results are shown in bold.

Method	Backbone	SSL	Input	$AP_{\text{BEV}}(\%)\uparrow$		$AP_{\text{3D}}(\%)\uparrow$	
				AP@0.3	AP@0.5	AP@0.3	AP@0.5
EchoFusion [7]	*ResNet-50	✗	2D (RA)	54.45	42.28	51.75	24.65
RTNH [2]	3D-SCB	✗	3D (RAE)	61.38	46.47	53.05	17.98
ViT-B [54]	ViT-B	✗	2D (RA)	60.77	45.30	55.24	27.45
MAE [17]	ViT-B	✓	2D (RA)	56.88	42.86	52.97	26.47
RadarMAE	*ViT-B	✓	2D (RA)	**64.17**	**48.44**	**58.85**	**30.93**

patches, observing a further slight performance increase. Motivated by these findings, we introduced our proposed axial tokenization, which significantly boosts the performance to the highest AP_{BEV} of 57.46%. This confirms that axial tokenization is highly meaningful for effectively recognizing axis-specific resolutions and preserving the spatial continuity of extended sinc-shaped sidelobes without fragmentation. Furthermore, decoupling the spatial axes drastically reduces the total number of tokens from 448 in the 8×8 configuration to 142. As a result, RadarMAE requires less than half the pre-training computational cost compared to the asymmetric MAE variant, while delivering superior object detection accuracy.

5.3 Comparison with Baseline Methods

In this setup, we conduct experiments on the K-Radar-20 subset to ensure a fair comparison with current radar object detection baselines. Additional results on the full K-Radar dataset are reported in Appendix D.

Standard MAE vs. RadarMAE. We further investigate the impact of directly applying a standard vision-centric MAE to radar data. As shown in Table 2, the standard MAE baseline consistently degrades performance across all spatial evaluation metrics compared to the ViT-B model trained from scratch without SSL. This notable drop indicates that naively transferring image-based masked modeling to radar perception is ineffective and can even be detrimental, primarily due to the substantial domain gap and distinct physical characteristics of radar signals. In contrast, RadarMAE successfully overcomes this limitation by explicitly incorporating radar-specific physical priors. Despite being built upon a standard ViT architecture, our proposed framework achieves the best overall performance, reaching a peak of 64.17% in AP_{BEV}@0.3 and 30.93% in AP_{3D}@0.5. These results demonstrate that injecting domain-aware physical priors during SSL is essential for learning effective and transferable radar representations.

Comparison with Other Radar Object Detection Methods. To further evaluate the effectiveness of RadarMAE for radar object detection, we compare it with representative radar detection baselines across different radar input modalities on the K-Radar-20 dataset, as shown in Table 2. Our primary baseline is the single-modality 2D RA configuration of EchoFusion [7], which serves as the most relevant supervised baseline since both models operate on identical input settings without camera, LiDAR, or temporal information. Compared to EchoFusion, RadarMAE achieves substantial absolute gains in both AP_{BEV}@0.3 and AP_{3D}@0.3. Given the identical input representations, this significant margin demonstrates the superior effectiveness of our SSL paradigm in learning robust radar feature representations directly from dense RA heatmaps. We further compare RadarMAE with RTNH [2], a widely used baseline but with 3D radar RAE tensor input, by employing the same anchor-based detection head. This comparison effectively isolates the impact of the radar input representation and the encoder. Despite relying solely on 2D RA heatmaps without explicit elevation information, RadarMAE yields superior performance over the 3D RAE-based RTNH. Most notably, our method achieves a dramatic absolute improvement in the stricter AP_{3D}@0.5 metric, nearly doubling the

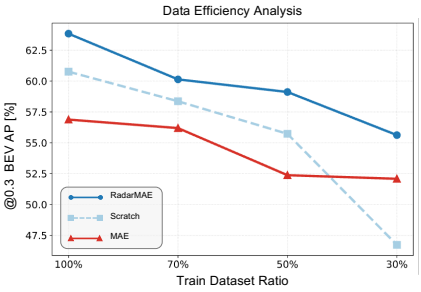


Figure 4: Data efficiency comparison using BEV AP@0.3.

localization accuracy of the baseline. Similar consistent improvements across other stringent metrics, such as $AP_{\mathrm{BEV}}@0.5$ and $AP_{\mathrm{3D}}@0.3$, further confirm that the proposed SSL framework learns highly robust and spatially consistent radar representations capable of precise 3D localization.

5.4 Ablation Studies

Comparison by Training Dataset Ratio. A primary motivation for adopting SSL is to improve data efficiency, which is particularly critical in the radar domain due to the immense difficulty and high cost of manual annotation. To thoroughly evaluate this, we conduct further experiments by varying the available training dataset ratio from 100% down to 30%, as illustrated in Fig. 4. Comparing the standard MAE with the model trained from scratch reveals a notable observation: when labeled data is extremely scarce, specifically at the 30% ratio, the standard MAE provides a distinct pre-training advantage over the scratch baseline. However, as the volume of training data increases, this benefit diminishes, and the standard MAE paradoxically underperforms the scratch model. This inversion implies that misaligned vision-centric priors in standard MAE ultimately hinder the network from fully leveraging larger amounts of radar data. Conversely, RadarMAE exhibits superior data efficiency across all scenarios. By correctly embedding radar-specific physical priors, RadarMAE not only prevents the degradation observed in standard MAE but also consistently maintains a significant performance gap over the scratch baseline across all data availability regimes.

Effect of Physical Distribution-Based Loss. To validate the efficacy of our proposed loss function, we compare different reconstruction losses in Table 3. The standard L2 loss (*i.e.,* MSE) yields the lowest performance (57.66% AP_{BEV}), empirically confirming our hypothesis that conventional MSE oversmooths distinct radar signatures by treating vital multipath tails and sinc-shaped sidelobes as uniform noise. Switching to an L1 objective mitigates this aggressive blurring effect, while applying an intensity-weighted strategy ($L1_w$) further compels the network to prioritize highly reflective target regions, boosting AP_{BEV} to 63.17%. Finally, integrating our full physical distribution-based loss ($\mathcal{L}_{\mathrm{total}}$) achieves the highest spatial localization accuracy with a 64.17% AP_{BEV}. By explicitly aligning the higher-order moments of the azimuth sidelobes and minimizing the 1D Wasserstein distance for the range multipath, $\mathcal{L}_{\mathrm{total}}$ successfully prevents the destruction of domain-specific geometric semantics. This substantial gain underscores that explicitly injecting structural statistical priors—rather than merely calculating pixel-wise distances—is critical for robust radar representation learning.

Table 3: Ablation study on reconstruction loss formulations. All metrics are evaluated at IoU 0.3.

Loss	$AP_{\mathrm{BEV}}(\%)\uparrow$
$L2$	57.66
$L1$	62.85
$L1_w$	63.17
$\mathcal{L}_{total}$	64.17

6 Conclusion

In this paper, we proposed **RadarMAE**, a radar-centric MAE framework tailored specifically to the physical properties of FMCW radar signals. While conventional MAEs rely on vision-oriented isotropic patching and pixel-wise reconstruction, RA heatmaps exhibit anisotropic sinc-shaped structures and axis-specific physical artifacts that require dedicated modeling. To bridge this domain gap, RadarMAE introduces an axial tokenization strategy that preserves range-azimuth resolution disparities and contiguous sinc-lobe geometry. Furthermore, by decoupling the reconstruction objectives of the two axes, our physical distribution-based loss explicitly captures heavy-tailed multipath reflections along the range axis and deterministic sidelobe spikes along the azimuth axis. Extensive experiments on the K-Radar benchmark using only single-frame RA maps demonstrate that injecting these domain-specific physical priors enables robust radar representation learning and substantially improves downstream object detection performance. These results highlight the importance of physically grounded self-supervised learning as a promising direction for radar perception.

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A Limitations and Future Works

Despite these promising results, RadarMAE currently focuses on learning radar-specific physical priors from 2D RA heatmaps. This choice provides a computationally efficient input representation, but it does not explicitly exploit elevation information. A natural future direction is to extend RadarMAE to higher-dimensional radar representations, such as RAE heatmaps. Such an extension could better capture vertical structures and object geometry that are not fully represented in 2D RA heatmaps, potentially leading to improved detection performance at the cost of increased computational complexity. Another limitation is that our evaluation is mainly centered on object detection. While object detection is a core downstream task for autonomous driving perception, the applicability of RadarMAE to other radar perception tasks remains underexplored. Future work should therefore evaluate the learned representation on broader downstream tasks, including semantic segmentation and object tracking. In addition, incorporating Doppler information is another promising direction for improving the discrimination of moving objects and stationary clutter, particularly when they exhibit similar intensity patterns. We believe these extensions will further improve the generality and robustness of radar representation learning across diverse real-world scenarios.

B Broader Impacts

This work may improve radar-based perception for autonomous driving, particularly under adverse weather and low-visibility conditions where camera- or LiDAR-based systems can degrade. By using self-supervised pre-training, RadarMAE may also reduce reliance on costly manual annotations for radar heatmaps. At the same time, radar perception is safety-critical. Improved benchmark performance does not guarantee safe deployment, and failures such as missed detections, false positives, or poor generalization could affect downstream driving decisions. Thus, RadarMAE should be viewed as a representation learning framework that requires extensive real-world validation before deployment. More capable radar sensing may also raise privacy and misuse concerns, since radar can reveal the presence, motion, and spatial behavior of people and vehicles. Responsible use should include appropriate data governance and compliance with relevant safety and privacy regulations. Finally, although the proposed 2D RA-based design is relatively efficient, future extensions to higher-dimensional radar inputs may increase computational cost, making energy efficiency and deployment practicality important considerations.

C Implementation Details

Pre-training. We pre-train a ViT-Base encoder with an embedding dimension of 768, 12 Transformer layers, 12 attention heads, and an MLP ratio of 4.0. The input RA maps are generated from K-Radar and resized to 108×258 (azimuth$\times$range) with a single channel. The radar intensity values are converted to the dB scale and normalized using dataset-level statistics. We adopt the proposed axial stripe tokenizer and use Physical Positional Encoding based on Fourier features. During pre-training, we apply intensity-weighted block masking with a masking ratio of 0.75. The model is optimized using a patch-intensity-weighted ℓ_1 reconstruction loss together with the proposed axis-specific distribution-matching losses for multipath and sidelobe structures, with both loss weights set to 0.1. Pre-training is performed for 800 epochs using AdamW [55] with a base learning rate of 1×10^{-5}, weight decay of 0.05, a 50-epoch warm-up, and a cosine learning-rate schedule.

Fine-tuning. For downstream 3D object detection, we initialize the encoder with the pre-trained RadarMAE weights and attach it to an RTNH-style anchor-based detection pipeline with polar-BEV feature refinement. The region of interest is set to $[0, 72] \times [-6.4, 6.4] \times [-2.0, 7.6]\,\mathrm{m}$, with a voxel size of $0.4\,\mathrm{m}$ and an azimuth field of view of $\pm 53^\circ$. We evaluate the *Sedan* class using K-Radar v1.0 annotations under the KITTI evaluation protocol. The detector is fine-tuned using AdamW with an initial learning rate of 1×10^{-4}, a minimum learning rate of 1×10^{-5}, weight decay of 10^{-3}, batch size 8, and cosine annealing for 11 epochs.

Compute resources. All experiments are conducted on a single workstation equipped with NVIDIA H200 GPUs (140 GB HBM3e per GPU). Pre-training of the ViT-Base RadarMAE model for 800 epochs on the full K-Radar pre-training set is performed on a single H200 GPU and takes approximately 9 hours, with a peak GPU memory consumption of about 17 GB. Fine-tuning the RTNH-style

detector for 11 epochs on a single H200 GPU takes approximately 0.5 hours per run and uses about 7 GB of GPU memory. The full set of experiments reported in the main paper and the supplementary material—including the pre-training ablations across tokenizer, positional encoding, masking, and loss configurations, the downstream fine-tuning runs across seeds and ablations, and downstream evaluation—amounts to roughly 200 GPU-hours on a single H200. Beyond the runs reported in the paper, the overall research project required additional compute for preliminary experiments, hyperparameter exploration (e.g., masking ratio, patch size, loss weights, learning-rate schedules), and design iterations on the tokenizer and positional encoding, which we estimate to be approximately 2–3$\times$ the compute of the final reported experiments.

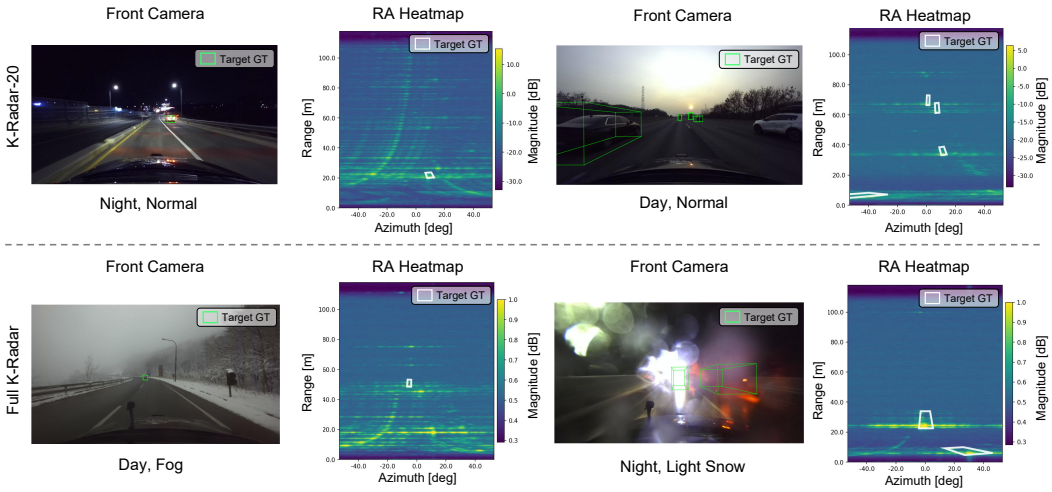


Figure 5: Qualitative examples of front-camera images and corresponding RA heatmaps from K-Radar-20 and the full K-Radar dataset. K-Radar-20 mainly contains normal-weather scenes, whereas the full K-Radar dataset includes more challenging adverse weather conditions such as fog and light snow. Green boxes indicate target ground-truth annotations in the camera view and their corresponding locations in the RA heatmaps.

D Performance on the Full K-Radar Dataset

Table 4: Comparison between K-Radar-20 and the full K-Radar dataset.

Group	Category	K-Radar-20		Full K-Radar	
		Seq.	Frames	Seq.	Frames
Scale	Total	20	13,008	58	34,994
Time	Day	13	9,008	35	21,920
	Night	7	4,000	23	13,074
Weather	Normal	19	12,781	19	12,781
	Overcast	1	227	2	825
	Rain	0	0	8	4,780
	Sleet	0	0	12	7,040
	Fog	0	0	6	3,578
	Light snow	0	0	4	2,237
	Heavy snow	0	0	7	3,753

To clarify the difference between K-Radar-20 and the full K-Radar dataset, we summarize their scale, time-of-day distribution, and weather diversity in Table 4. K-Radar-20 contains 20 sequences and 13,008 frames, most of which are collected under normal weather conditions. In contrast, the full K-Radar dataset contains 58 sequences and 34,994 frames, covering diverse adverse weather

conditions such as rain, sleet, fog, light snow, and heavy snow [2]. Fig. 5 provides qualitative examples showing that the full dataset includes more challenging weather scenes where camera observations are degraded while RA heatmaps remain informative.

We further evaluate RadarMAE on the full K-Radar dataset to verify its scalability under broader data distributions. As shown in Table 5, RadarMAE achieves the best performance across all BEV and 3D detection metrics. Notably, it outperforms both the scratch ViT-B and standard MAE baselines, confirming that radar-specific physical priors remain effective beyond the K-Radar-20 subset. RadarMAE also surpasses RTNH despite using only 2D RA maps instead of 3D RAE inputs, further demonstrating the effectiveness of physically grounded representation learning for robust radar object detection.

Table 5: Performance comparison on the full K-Radar dataset. Best results are shown in bold.

Method	Backbone	SSL	Input	$AP_{\text{BEV}}(\%)\uparrow$		$AP_{\text{3D}}(\%)\uparrow$	
				AP@0.3	AP@0.5	AP@0.3	AP@0.5
RTNH [2]	3D-SCB	✗	3D (RAE)	58.04	42.60	49.65	17.87
ViT-B [54]	ViT-B	✗	2D (RA)	58.45	44.18	53.29	28.99
MAE [17]	ViT-B	✓	2D (RA)	55.32	42.85	51.99	28.00
RadarMAE	*ViT-B	✓	2D (RA)	**60.48**	**47.76**	**56.43**	**29.04**

E Qualitative Results

E.1 Object Detection Results

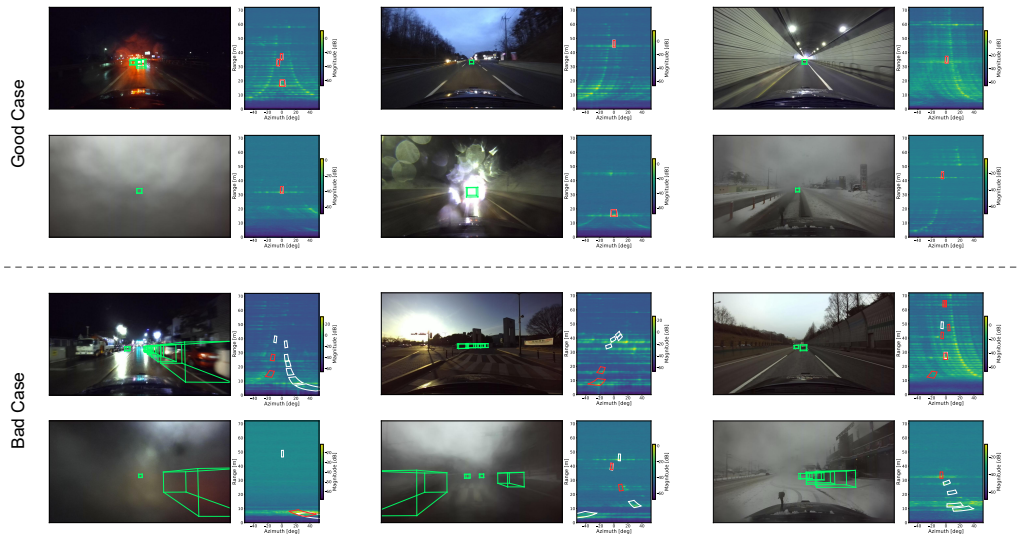


Figure 6: Qualitative detection results of RadarMAE on challenging driving scenes. The top group shows representative good cases, where RadarMAE accurately localizes target vehicles from single-frame RA heatmaps despite adverse visual conditions such as nighttime, fog, tunnels, and snowy roads. The bottom group shows representative bad cases, where weak target responses, severe clutter, or ambiguous radar reflections lead to missed detections or inaccurate localization. In the camera view, green boxes denote ground-truth annotations. In the RA heatmaps, white boxes denote ground-truth target regions, while red boxes denote the predictions of RadarMAE.

E.2 Visualization of Attention Maps for the Proposed Method.

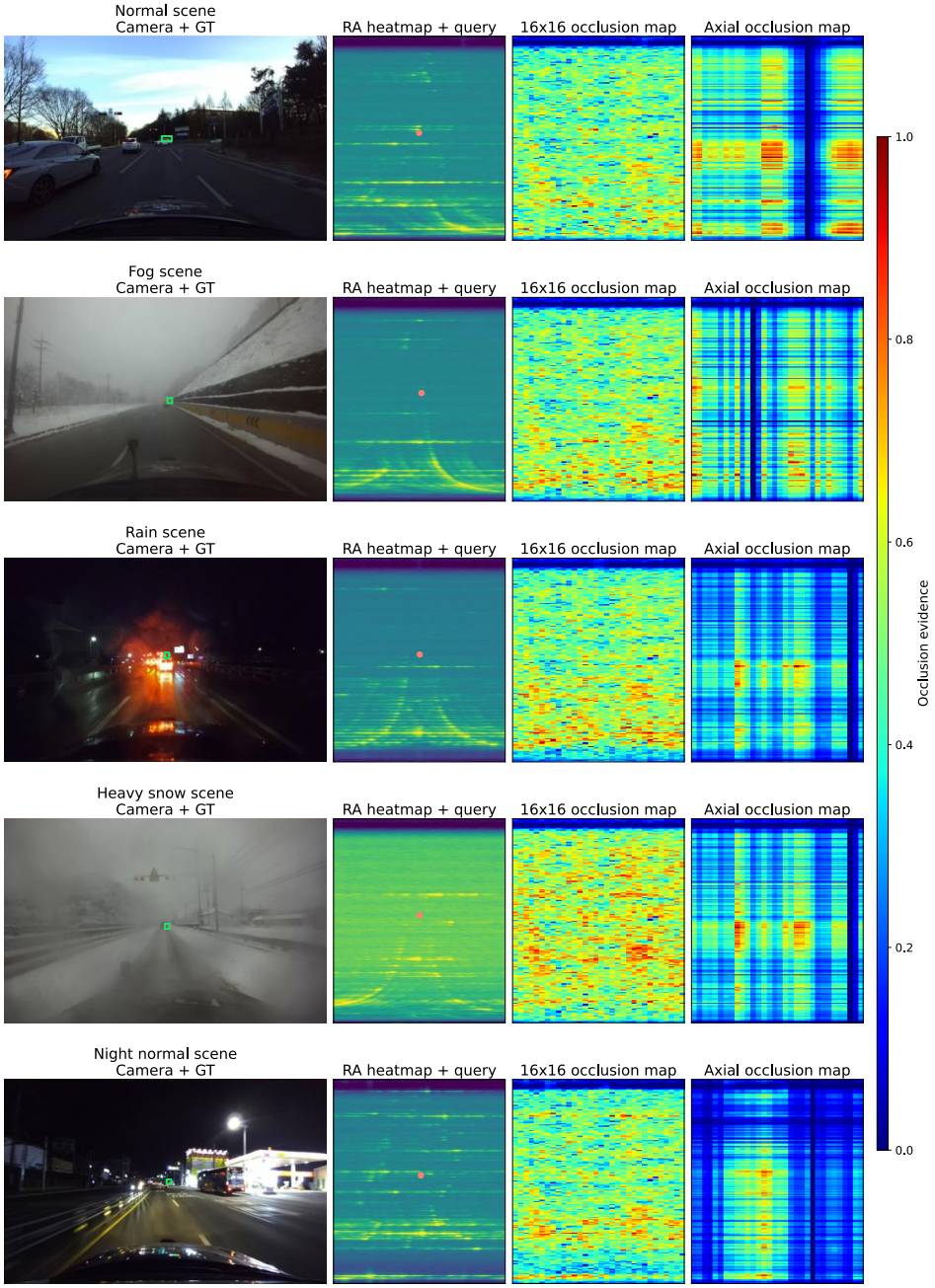


Figure 7: Visualization of occlusion-based attention evidence under diverse driving conditions. Each row shows a front-camera image with the ground-truth target, the corresponding RA heatmap with the query location, the occlusion map obtained from conventional 16×16 square patches, and the occlusion map obtained from the proposed axial tokenization. Compared with the noisy and spatially fragmented responses of square patch occlusion, axial occlusion produces more structured axis-aligned evidence around target-related radar signatures. This indicates that the proposed axial tokenization better preserves the continuous range and azimuth-wise physical structures of RA heatmaps across normal, fog, rain, snow and nighttime scenes.

F Weighted Block Masking

Fig. 8 illustrates the proposed weighted block masking strategy, where contiguous high-intensity regions in the RA heatmap are masked while a hint token is retained for stable reconstruction. As shown in Table 6, weighted block masking consistently outperforms random masking across all BEV and 3D detection metrics. In particular, it improves $AP_{\mathrm{BEV}}@0.3$ from 60.95% to 64.17% and $AP_{\mathrm{3D}}@0.5$ from 28.22% to 30.93%. These results indicate that explicitly masking informative target regions better encourages the model to recover meaningful radar signatures, rather than relying on trivial background interpolation.

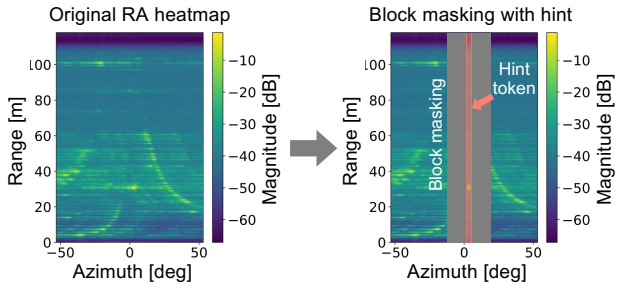


Figure 8: Proposed intensity-weighted block masking. Contiguous segments of $5 \sim 8$ tokens are masked to hide sparse targets and reflect axial artifact dispersion. To prevent information loss and stabilize optimization, at least one unmasked anchor token per block is preserved by a visible hint mechanism.

Table 6: Comparison of masking strategies on K-Radar-20. BEV and 3D AP are reported at IoU 0.3/0.5. Best results are shown in bold.

Masking	$AP_{\mathrm{BEV}}(\%)\uparrow$		$AP_{\mathrm{3D}}(\%)\uparrow$	
	$AP@0.3$	$AP@0.5$	$AP@0.3$	$AP@0.5$
Random	60.95	44.98	55.54	28.22
Weighted block	**64.17**	**48.44**	**58.85**	**30.93**

G Positional Embedding

We further analyze the effect of positional encoding by comparing conventional positional embedding (PE) with the proposed Physical Positional Encoding (PPE). In this experiment, all components of the final RadarMAE framework are kept fixed, and only the positional encoding strategy is changed. PPE replaces index-based embeddings with Fourier features computed from absolute radar sensor coordinates, namely range in meters and azimuth in radians, which are then projected to the hidden dimension through an MLP and added to the axial tokens with learnable axis-type embeddings. This allows the model to encode physically meaningful geometry rather than relying only on discrete token indices. As shown in Table 7, PPE consistently improves both BEV and 3D detection performance, demonstrating that radar-aware positional encoding provides more informative spatial cues for robust downstream localization.

Table 7: Comparison of positional embedding strategies on K-Radar-20. BEV and 3D AP are reported at IoU 0.3/0.5. Best results are shown in bold.

Positional Embedding	$AP_{\mathrm{BEV}}(\%)\uparrow$		$AP_{\mathrm{3D}}(\%)\uparrow$	
	$AP@0.3$	$AP@0.5$	$AP@0.3$	$AP@0.5$
PE	61.23	45.04	56.19	28.81
PPE	**64.17**	**48.44**	**58.85**	**30.93**

H Fine-Tuning Details

Feature Reshaping. After the self-supervised pre-training, we discard the decoder and initialize the encoder with the learned weights to adapt the representations for the downstream detection task. To construct a dense spatial feature map from the axial tokens, we employ a direct stripe-to-BEV mapping that projects the axial token bank onto a 2D BEV grid. During this projection, geometric alignment is achieved by generating polar reference points based on the BEV query coordinates, where learned offsets explicitly compensate for the nonlinear coordinate transformations between the radar polar space and the Cartesian BEV space. Finally, a lightweight ASPP-Lite block [52] is applied to the projected BEV feature map to enrich multi-scale contextual information without heavy computational overhead.

Detection Head. To ensure that any improvements in detection performance are strictly attributable to our proposed representation learning framework, we adopt a standard anchor-based detection head identical to the RTNH baseline [2]. The entire network is fine-tuned end-to-end using a joint detection loss $\mathcal{L}_{\text{det}}$, which is a combination of Focal loss for classification, smooth-L1 loss for bounding box regression, and a directional classification loss.